

Appendix to: A ProbLog program to infer individual genotypes from familial phenotypes in autosomal, X-linked, and Y-linked Mendelian disorders

Maxime Mahout  ^{1*}

¹ Inria Saclay, EPI Lifeware, 91120, Palaiseau, France.

* Correspondence: maxime.mahout@inria.fr

EXAMPLES FROM BENETT'S BOOK

The Johnson Family

Bob Johnson is an intellectually delayed patient affected with proteinuria. He was born blind and he had surgeries for glaucoma and cataracts. His family is seeking a genetic explanation for his characteristics. The full pedigree of Bob is presented in Figure 1.

Bob's mother, Ann, underwent surgery for cataracts, but is otherwise unaffected. The genealogy of Bob's father, Jim, is unknown. Bob's sister Julie is also unaffected by the suspected intellectual disorder. On the other hand, Ann's now deceased brother John seems to have presented the same symptoms. As well, Sam, the son of a female cousin of Ann, from the family of Ann's mom, seems to have been affected by this disorder. No genotyping was performed.

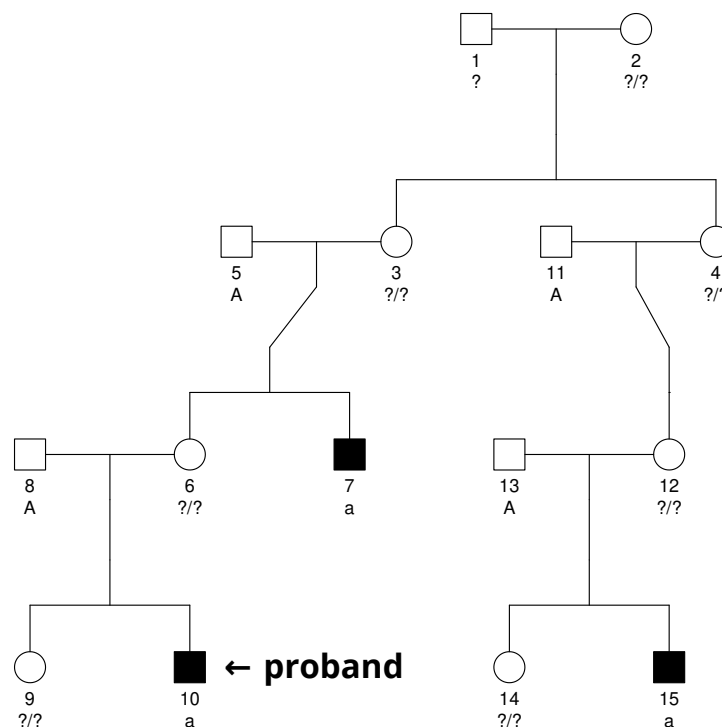


Figure 1: Pedigree of Bob, patient to be diagnosed with X-linked oculocerebrorenal syndrome. Phenotypes marked as affected (filled black), unknown or unaffected (filled white). Pedigree constructed with *pedtools* from *pedsuite*.

Using our *mendelprob.pl* method, we can perform probability analyzes for the direct pedigrees of Bob, John, and Sam, and try to conjecture the disorder. Let us first try the dominant transmission hypothesis, where *A* is the disease-conferring allele, on Bob's *direct pedigree* up to his grandparents (individuals no. 3, 5):

```
prevalence("a", 999/1000). lastgen(2).
evidence(not m_show(0, "A")). % Father of Ann
evidence(not f_show(0, "A")). % Mother of Ann
evidence(not m_show(1, "A")). % Jim
evidence(not f_show(1, "A")). % Ann
evidence(f_family_descendant(1)). % Ann
evidence(m_family_descendant(2)). % Bob
evidence(m_show(2, "A")). % Bob
```

This code returns inconsistent evidence when in the logic program evidence evaluation. This means the dominance hypothesis is in direct contradiction with the phenotypes shown. We can therefore reliably exclude that hypothesis. Now for the same code but with the recessive hypothesis:

```
prevalence("a", 1/1000). lastgen(2).
evidence(not m_show(0, "a")). % Father of Ann
evidence(not f_show(0, "a")). % Mother of Ann
evidence(not m_show(1, "a")). % Jim
evidence(not f_show(1, "a")). % Ann
evidence(f_family_descendant(1)). % Ann
evidence(m_family_descendant(2)). % Bob
evidence(m_show(2, "a")). % Bob
```

This time the evidence is consistent. The particularity of this example is that from these phenotypes alone, no conclusion can be taken on whether the disorder is autosomal or X-linked. However, we observe that as the prevalence of *a* increases, the probability of the disorder being X-linked gets closer and closer to 1. For instance, with a prevalence of $1/2$, we get `0.5:disease("auto")` and `0.5:disease("x")`. For $p = 1/10$, we get `1/6:disease("auto")`, `5/6:disease("x")`. For $p = 1/1000$, we get around `0.002:disease("auto")` and `0.998:disease("x")`. Among the many probabilities calculated, one of them is conclusive: `1:f_carry(1, "Aa")` – Bob's mom, Ann, must carry the genotype *Aa*. However, in both X-linked and autosomal dispositions, that genotype wouldn't make sense. This is a first step towards the diagnosis.

Previously, we weren't able to derive anything from the siblings. This is because the probabilities of inheriting certain genotypes from the parents do not change between one sibling and another. In addition to the current direct pedigree of Bob (3,5; 6,8; 10 – in [Figure 1](#)), we could calculate the probabilities for the direct pedigree of Sam (4, 11; 12,13; 15), Bob's affected cousin. Unfortunately, for Sam we would encode exactly the same evidences and this would bring no new information. But what if it's across different generations, like with the case of Bob's uncle John (3, 5; 7). We write the following evidences:

```
prevalence("a", 1/1000). lastgen(1).
evidence(not m_show(0, "a")). % Father of John
evidence(not f_show(0, "a")). % Mother of John
evidence(m_show(1, "a")). % John
evidence(m_family_descendant(1)). % John
```

This time we derive the atom `1:f_carry(0, "Aa")`, meaning that John's mother and Bob's grandmother must have necessarily had the genotype "Aa". Upon reinserting the atom as an evidence `evidence(f_carry(0, "Aa"))` into Bob's pedigree, we get stronger support for the X-linked disorder hypothesis, with `0.999:disease("x")` against `0.001:disease("auto")`. This is understood intuitively, here only the males are affected and X-linked disorders affect males more strongly due to them only having one X copy.

In actuality, Bennett reports that Bob Johnson's disorder should be Lowe syndrome, or X-linked oculocerebrorenal syndrome. Although with *mendelprob.pl* we can calculate probabilities of the disorder being either autosomal or X-linked, let us assume it is X-linked for certain, using `evidence(disease("x"))`, in order to compare our tool with *pedprobr*. While *pedprobr* does not allow for phenotype reasoning and uncertain reasoning over disorder types between autosomal, X-linked and Y-linked, we are thankfully able to calculate X-linked inheritance. The key is like in the previous *pedprobr* example to specify as most genotypes as desirable, as is done in [Figure 1](#).

We are now interested in the great-grandparents of Bob (no. 1, 2). Specifying the genotypes of all males depending on their visible phenotype, and specifying the genotype of Bob's grandmother as *Aa* (no. 3), we get the following probabilities for the genotypes of great-grandparents (no. 1, 2) with both *mendelprob.pl* and *pedprobr*: `0.0005:f_carry(0,"aa"), 0.4995:f_carry(0,"AA"), 0.5:f_carry(0,"Aa"), 0.5:m_carry(0,"A-"), 0.5:m_carry(0,"a-")`. An interesting observation is this means that there is a non-negligible chance that Bob's great maternal grandfather displayed the disease's phenotype, yet was able to have two unaffected daughters. This information will be useful to Bob's family.

Here is the final *mendelprob.pl* code for the analysis of the four generations:

```
prevalence("a", 1/1000). lastgen(3).
evidence(disease("x")). % Lowe disorder
evidence(f_carry(1, "Aa")). % Greatmother of Ann
evidence(m_carry(1, "A-")). % Greatfather of Ann
evidence(m_carry(2, "A-")). % Jim / Father of Bob
evidence(f_carry(1, "Aa")). % Mother of Ann
evidence(f_family_descendant(1)). % Ann
evidence(f_family_descendant(2)). % Ann
evidence(m_family_descendant(3)). % Bob
evidence(m_carry(3, "a-")). % Bob
```

REFERENCES

- [1] Robin L. Bennett. *The Practical Guide to the Genetic Family History*. John Wiley & Sons, September 2011. ISBN 978-1-118-20981-3.